DUYGU KAYTAN

Robotics & Autonomous Systems Researcher

Cyber-Physical Systems | Trustworthy Autonomy | Control & AI | Embedded Robotics

PROFILE



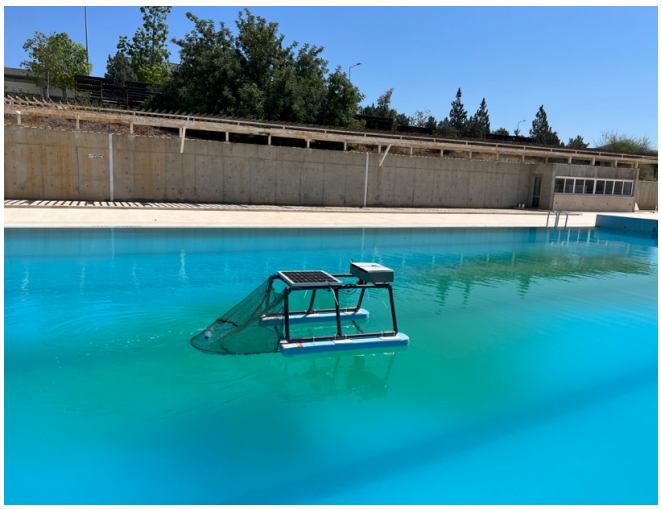
MSc researcher in Intelligent Transportation Systems with a background in Electrical and Electronics Engineering. My work focuses on trustworthy and resilient autonomous robotics, combining state estimation, sensor fusion, anomaly detection, interpretable AI, and safety-oriented decision architectures.

I develop and experimentally evaluate algorithms using independently assembled mobile robotic platforms and embedded sensing systems. My research combines algorithmic design, quantitative evaluation, and physical system validation.

PROJECTS

Autonomous Solar-Powered Surface Cleaning Robot

Multidisciplinary Engineering Project (ESC 491–492) | Middle East Technical University (METU)



Designed and built an autonomous solar-powered surface vehicle intended to collect floating waste from marinas and coastal waters. The project integrated mechanical design, propulsion, onboard electronics, solar energy management, and autonomous operation.

Developed within a multidisciplinary engineering team, with an emphasis on complete hardware realization, system integration, and physical testing.

Key Contributions

- Designed and implemented the physical platform and complete electrical/electronic hardware architecture, including propulsion and power distribution.
- Developed the solar-energy subsystem and integrated onboard control hardware, Raspberry Pi, Pixhawk, GPS, and telemetry.
- Led system integration and physical testing; collaborated with computer engineering teammates on autonomous navigation, control-parameter calibration, and mission planning.





Project demonstration: https://www.youtube.com/watch?v=cjh0bYrHCVc

EXPERIMENTAL ROBOTICS & EMBEDDED SYSTEMS DEVELOPMENT

Independent Experimental Research | Personal Robotics Laboratory | Ongoing



I build and test autonomous mobile robotic platforms in an independently assembled experimental laboratory. My work emphasizes hardware–software integration, heterogeneous sensing, controlled disturbances, and dependable robot behavior under uncertainty.

Platforms include small mobile rovers and embedded sensing systems supporting research in state estimation, anomaly detection, and reactive navigation.



Key Areas of Work

- Mobile robotic platform development and system integration
- LiDAR, IMU, wheel encoders, GPS/GNSS, and electrical sensing
- Python and MATLAB/Simulink data analysis and experimental evaluation
- Raspberry Pi, ESP32, and Teensy 4.1 integration
- Fault injection, anomaly monitoring, and physical validation
- Motor control, power electronics, instrumentation, soldering, and debugging

SELECTED MULTIDISCIPLINARY PROJECT

TEKNOFEST 2022 — Combat UAV Development

ALYA Team | Middle East Technical University | Fighting UAV Category

Contributed to a multidisciplinary autonomous UAV project integrating flight control, onboard computation, telemetry, and ground-control interaction. The team brought together electrical/electronics, computing, mechanical, and aerospace contributions.

Key Contributions

- Took responsibility for onboard electrical/electronic integration and associated hardware testing.
- Integrated Pixhawk-based flight-control hardware, Raspberry Pi onboard computing, power and telemetry subsystems.
- Performed hardware-level checks and calibration in coordination with software and airframe-development teammates.

This engineering experience complements my later research by grounding autonomy, embedded computing, and sensing in physical system constraints.

RESEARCH & PUBLICATIONS

My research develops a connected trajectory from adaptive state estimation and interpretable anomaly detection to active physical sensing and recurrent safe decision-making. Published work is distinguished below from manuscripts currently under editorial consideration.

PUBLISHED RESEARCH

Adaptive and Interpretable Reinforcement Learning-Enhanced Unscented Kalman Filtering for Autonomous Driving State Estimation

Duygu Kaytan | Applied Soft Computing (Elsevier) | Published, 2026

Proposed an interpretable reinforcement-learning-enhanced Unscented Kalman Filter in which structured agent actions adapt process and measurement noise covariances while retaining the physical meaning of classical filtering. The study evaluates nonlinear vehicle state estimation across changing motion and noise conditions in Python simulation, including repeated Monte Carlo experiments and statistical comparisons.

DOI: 10.1016/j.asoc.2026.115924

Explainable IMU Anomaly Detection: Impact of Window Size on Machine Learning Performance under Noise Injection

Duygu Kaytan and Tülay Yıldırım | SIU 2026 / IEEE Xplore | Published, 2026

Investigated machine-learning-based anomaly detection using IMU time-series data collected from a mobile robot under controlled noise injection. Compared temporal window sizes and classification approaches, with SHAP analysis to interpret model decisions and identify influential signal features.

DOI: 10.1109/SIU71813.2026.11636468

Intelligent Estimators: Reinforcement Learning Enhanced Unscented Kalman Filter for Autonomous Driving

Duygu Kaytan | 26th National Conference on Automatic Control (TOK 2025) | Published in conference proceedings, 2025

Presented the initial MATLAB/Simulink implementation of an RL-enhanced UKF for nonlinear vehicle state estimation. This conference contribution established an early prototype subsequently expanded into the journal-level study.

Proceedings DOI (whole volume): 10.5281/zenodo.17737744

Individual contribution:
https://www.researchgate.net/publication/408251867_Intelligent_Estimators_Reinforcement_Learning_Enhanced_Unscented_Kalman_Filter_for_Autonomous_Driving

MANUSCRIPTS UNDER EDITORIAL CONSIDERATION

Stop-Signal Bee: A Recurrent Behavioural Decision Architecture

Duygu Kaytan | Robotics and Autonomous Systems (Elsevier) | Submitted 2 August 2026; With Editor

Introduces Stop-Signal Bee (SSB), a recurrent behavioral decision architecture inspired by inhibitory signaling in collective honeybee decisions. Its modular design separates a reusable decision core from application-specific sensing and action interfaces. The manuscript examines reactive autonomous navigation through comparative simulation and physical mobile-robot validation.

Active Physical Watermarking Across Heterogeneous Sensing Modalities for Mobile Robot Anomaly Detection

Duygu Kaytan | Engineering Applications of Artificial Intelligence (Elsevier) | Submitted 6 October 2026

Investigates an active physical-watermarking approach that applies bounded actuator-level excitation and observes robot responses across electrical, inertial, and kinematic sensing modalities. The study uses physical mobile-robot experiments to assess whether controlled excitation provides complementary evidence for interpretable anomaly detection.

MASTER'S THESIS — ONGOING

An Explainable Trust-Aware Decision Framework for Anomaly Detection and Resilient Autonomous Robot Navigation

Yıldız Technical University | Intelligent Transportation Systems | Ongoing

The proposed thesis integrates anomaly detection, uncertainty-aware evaluation of information reliability, and two SSB-based decision layers addressing which information sources to trust and how the robot should respond. An independent runtime safety-monitoring layer is also planned. The research design includes separate detection and decision benchmarks, ablation, sensitivity and robustness analyses, computation-time assessment, and physical validation on autonomous mobile robots.

FUTURE RESEARCH DIRECTIONS

My ongoing research aims to advance trustworthy and resilient autonomous systems through interpretable state estimation, anomaly-aware sensing, and safety-oriented decision-making.

Building on my published reinforcement learning-enhanced UKF framework, I plan to investigate the integration of the Stop-Signal Bee (SSB) architecture into adaptive state estimation (SSB-UKF).

My longer-term research interests focus on extending SSB-inspired mechanisms to new robotic platforms and decision problems, particularly anomaly detection, distributed trust-aware decision-making in multi-agent and swarm robotic systems, and formal verification and runtime safety assurance of autonomous decisions.

TECHNICAL EXPERTISE

Robotics, Control & Cyber–Physical Systems

- Autonomous mobile robotic platforms; system integration and reactive navigation
- State estimation, Kalman filtering and UKF; sensor fusion
- Fault/anomaly-aware sensing and safety-oriented decision systems

AI, Modeling & Experimental Evaluation

- Machine learning, reinforcement learning, interpretable learning and explainable AI (SHAP)
- Anomaly detection; structured fault injection; physical-watermarking experiments
- Simulation, quantitative benchmarking, ablation/sensitivity studies, and statistical evaluation

Embedded Systems & Electronics

- Raspberry Pi, ESP32, Teensy 4.1, embedded sensing and control
- LiDAR, IMU, wheel encoders, GPS/GNSS, motor drivers, telemetry and power electronics
- Circuit assembly, soldering, system troubleshooting, instrumentation and prototyping

Programming & Software Tools

Python | MATLAB/Simulink | C/C++ | Git/GitHub | Introductory ROS

SELECTED LINKS

GitHub: https://github.com/KaytanTR

Research portfolio: https://github.com/KaytanTR/robotics-portfolio

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